

APOLLO 15

**CSM
RESCUE BOOK**

PART NO.

S/N

SKB32100115-354

1002

APOLLO 15
CSM RESCUE BOOK

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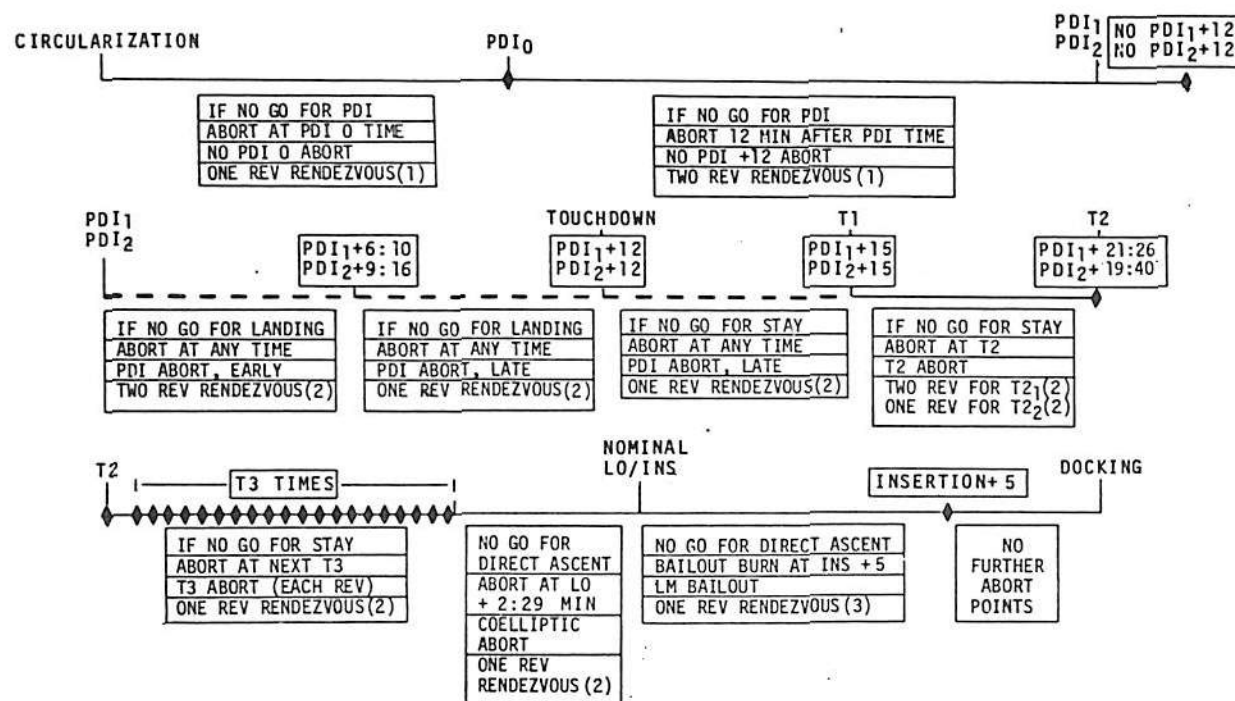
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LM ABORT POINTS



CSM RESCUE CASES

IF THE CSM BECOMES ACTIVE AFTER:

- (1) AN ACCURATE LM ABORT INITIATION MANEUVER COMPLETE THE ONE REV OR TWO REV RENDEZVOUS
A PARTIAL LM ABORT INITIATION MANEUVER INITIATE THE RESCUE TWO RENDEZVOUS
- (2) THE LM INSERTS INTO THE DESIRED ORBIT COMPLETE THE EARLY, ONE REV, OR TWO REV
RENDEZVOUS
THE LM INSERTS WITH AN UNDERSPEED OR OVERSPEED INITIATE A CSM BAILOUT, MANUAL INSERTION,
CSM DWELL, OR RESCUE TWO RNDZ AS DIRECTED
- (3) AN ACCURATE LM BAILOUT MANEUVER COMPLETE THE ONE REV RENDEZVOUS
A PARTIAL LM BAILOUT MANEUVER PERFORM THE P30 CSM BAILOUT BURN AND
COMPLETE THE ONE REV RENDEZVOUS

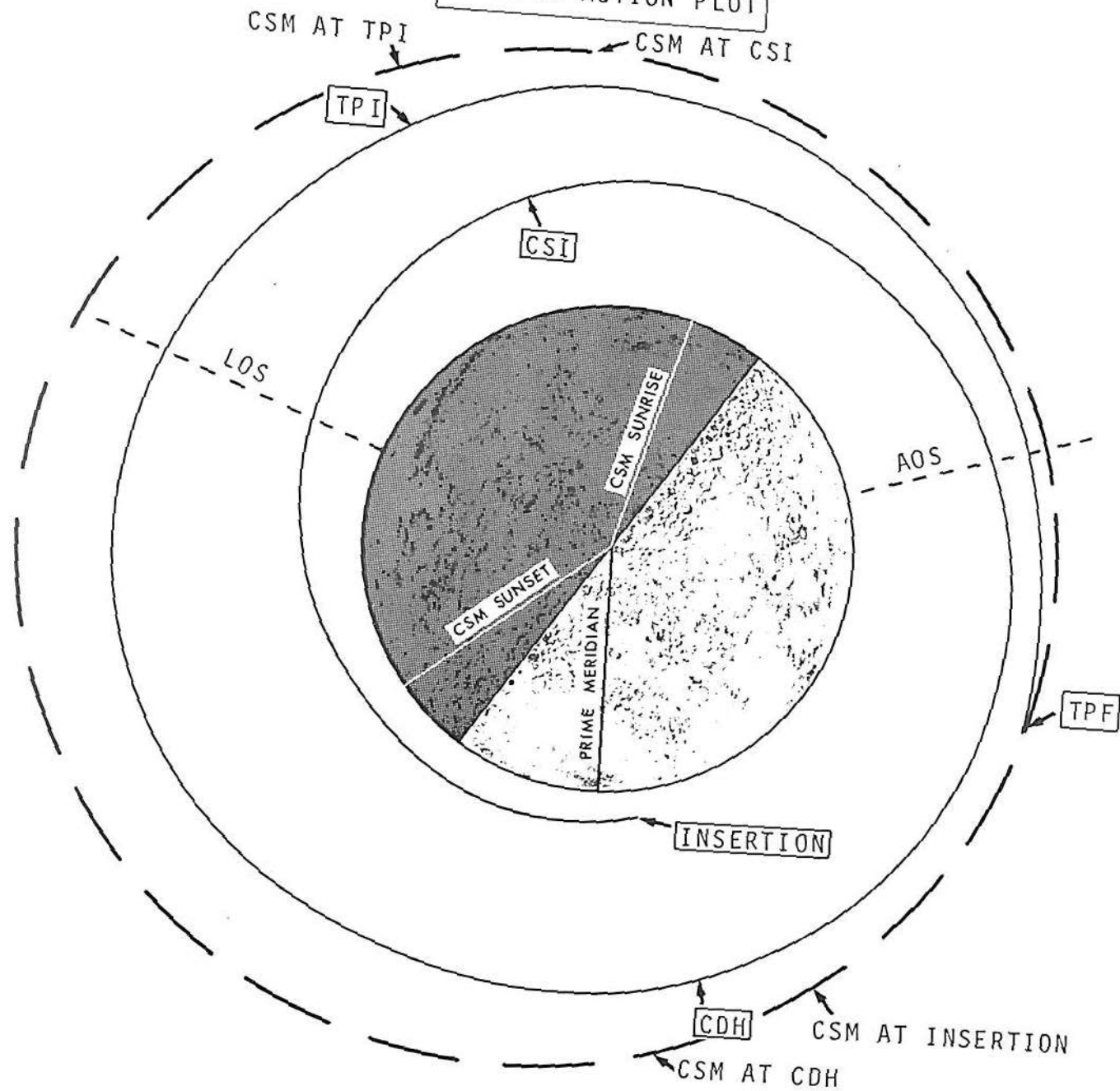
REQUIREMENTS FOR THE RESCUE BOOK
TO CONTAIN SIM POWER DOWN PROCEDURES
HAVE BEEN DELETED.

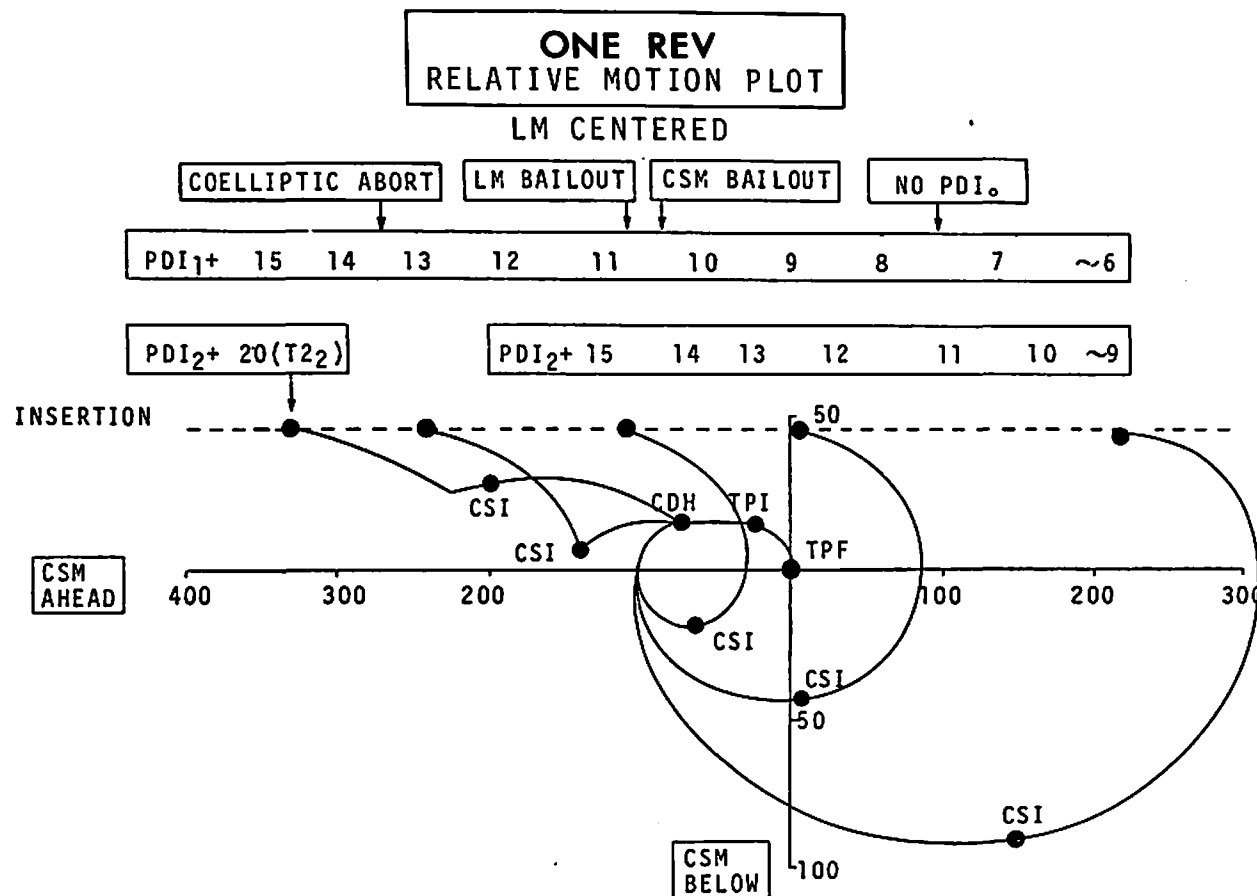
THESE PROCEDURES ARE NOW CONTAINED
ON PAGE S/1-40 OF THE CSM
SYSTEMS CHECKLIST.

REQUIREMENTS FOR THE RESCUE BOOK
TO CONTAIN SIM POWER DOWN PROCEDURES
HAVE BEEN DELETED.

THESE PROCEDURES ARE NOW CONTAINED
ON PAGE S/1-40 OF THE CSM
SYSTEMS CHECKLIST.

ONE REV
ORBITAL MOTION PLOT





ONE REV
BURN TIMES

NO PDI 0 ABORT
CSI AT A.I.M. +60 MIN
TPI AT 105:21:46.5

PDI 2, LATE ABORT
CSI AT INS +55 MIN
TPI AT 109:18:38.3

BACKUP COELLIPTIC
CSI AT INS +55 MIN
TPI AT 174:29:02.9

CSM BAILOUT
CSI AT INS +55 MIN
TPI AT 174:29:02.9

PDI 1, LATE ABORT
CSI AT INS +55 MIN
TPI AT 107:20:12.5

T2 ABORT, 2ND OPP
CSI AT INS +50 MIN
TPI AT 109:18:38.3

LM BAILOUT
CSI AT INS +55
TPI AT 174:29:02.9

ONE REV
TIMELINEPOWER DOWN SIM(SEE SYSTEMS CHECKLIST, PAGE S/1-40)
PCO; V48(11102,X1111)

INSERTION OR A.I.M.

MSFN UPLINK: LM S.V.

OBTAIN LM CSI AND TPI GETI'S (COPY)

P32 (SEE INPUT)

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

RECYCLE(COPY)

FINAL COMP(COPY); OBTAIN LM CSI DV(COPY); P40m

CSI BURN

P76m; OBTAIN LM PC GETI(COPY); P36m WITH LM GETI

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

FINAL COMP (COPY) TO BYPASS P52m AND P4Xm, ZERO DVY

OBTAIN LM PC DV (COPY)

P52m; P4Xm

TO BYPASS GYRO TORQUE: ENTER ON F 50 25
TO GYRO TORQUE: CMC MODE-FREE
PRO ON F 50 25

PLANE CHANGE

P76m; OBTAIN LM CDH GETI (COPY); P33m WITH LM GETI

RECYCLE(COPY)

FINAL COMP(COPY); OBTAIN LM CDH DV(COPY); P40m

CDH BURN

P76m

INSERTION		LM INSERTION TIME		
A.I.M. P76	33	LM A.I.M.GETI	:	:
	84	LM A.I.M.DV-LV	:	:

CSI COMPUTATION				
INPUT	11	LM CSI GETI		
	55	N +00001	ELEVATION +208.30	180° OPT +130.00
	37	LM TPI GETI		
RECYCLE	90	Y	Y DOT	LM Y DOT
	81	CSI DV-LV	.	+0000.0
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	CSI DV-LV	.	+0000.0
P76	84	LM CSI DV-LV	.	.

PLANE CHANGE COMPUTATION				
INPUT	33	LM PC GETI		
	90	Y	Y DOT	LM Y DOT
	81	PC DV-LV	.	+0000.0
P76	84	LM PC DV-LV	.	+0000.0

CDH COMPUTATION				
INPUT	13	LM CDH GETI		
	90	Y	Y DOT	LM Y DOT
	81	CDH DV-LV	.	.
RECYCLE	90	Y	Y DOT	LM Y DOT
	81	CDH DV-LV	.	.
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	CDH DV-LV	.	.
P76	84	LM CDH DV-LV	.	.

P34m, OVERWRITE N55 R2.

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

IF ONLY ONE SENSOR OPERATIONAL:
V57(00004, 00001)

-20
-12
0
+12
+15
+27
+30
+42

S
X
T
A
N
D
V
H
F

RECYCLE(COPY)

OBTAIN LM TPI GETI AND DV(COPY); FINAL
COMP, OVERWRITE N37 WITH LM TPI GETI

P40m

TPI BURN

S
/
V

P76m; P35m

FINAL COMP(COPY); OBTAIN LM MCC1 DV(COPY); P41m

MCC1

S
/
V

P76m; P35m

FINAL COMP(COPY); OBTAIN LM MCC2 DV(COPY); P41m

MCC2

GO TO TPF ACTIVITIES(PAGE 38)

TPF

TPI-ELEVATION OPTION

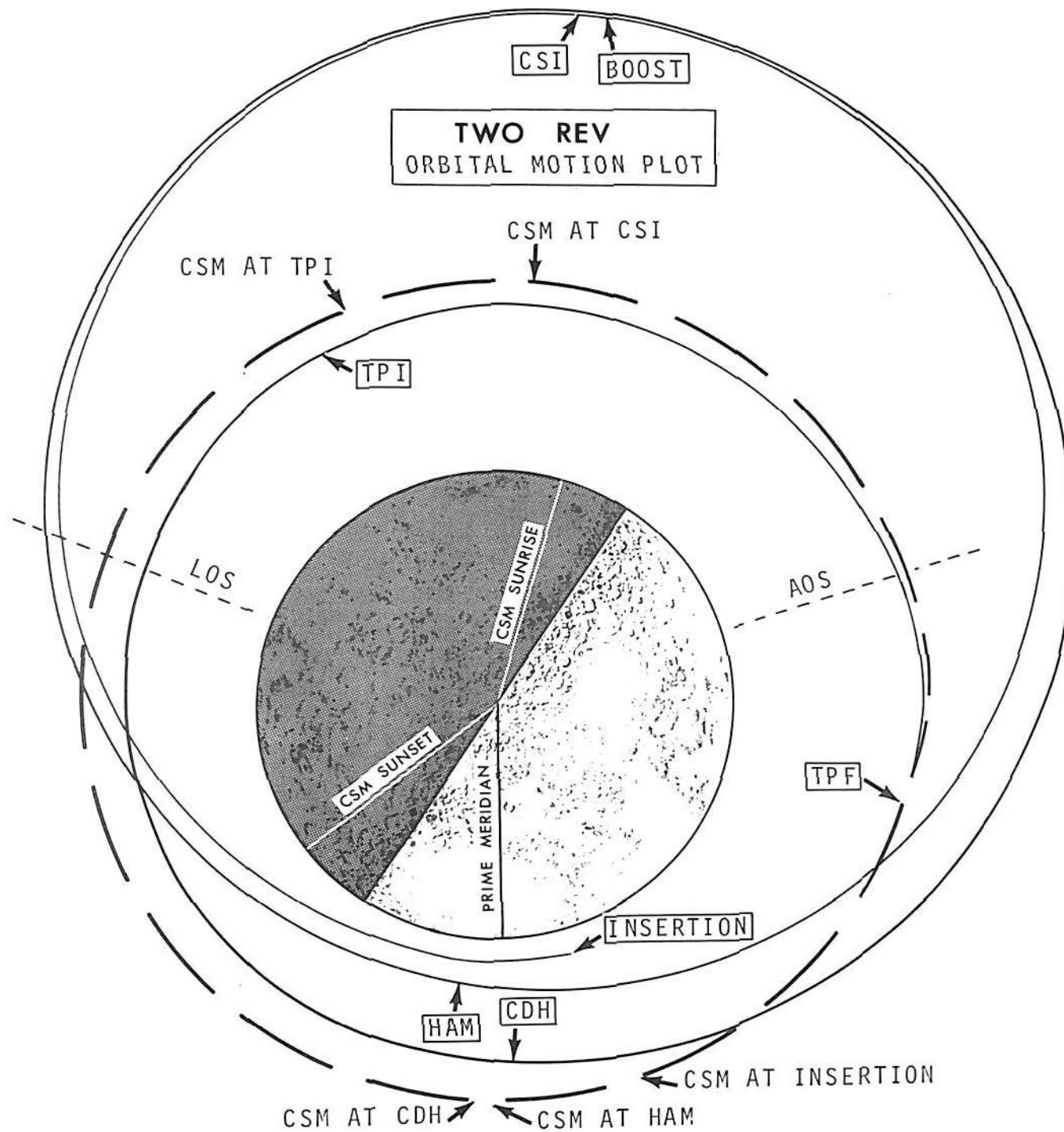
INPUT		TPI GETI		
		INTEG OPT	ELEVATION	TRANSFER
RECYCLE	37	PROCEED ON DISPLAYED GETI		
	55	+00000	+208.30	+130.00
	81			
FINAL COMP	37	PROCEED ON DISPLAYED GETI		
	55	+00000		+130.00
	81			
P76	33	LM TPI GETI		
	84	LM TPI DV-LV		

MCC COMPUTATION

		MCC1 DV-LV		
FIN COMP	81			
P76	84			

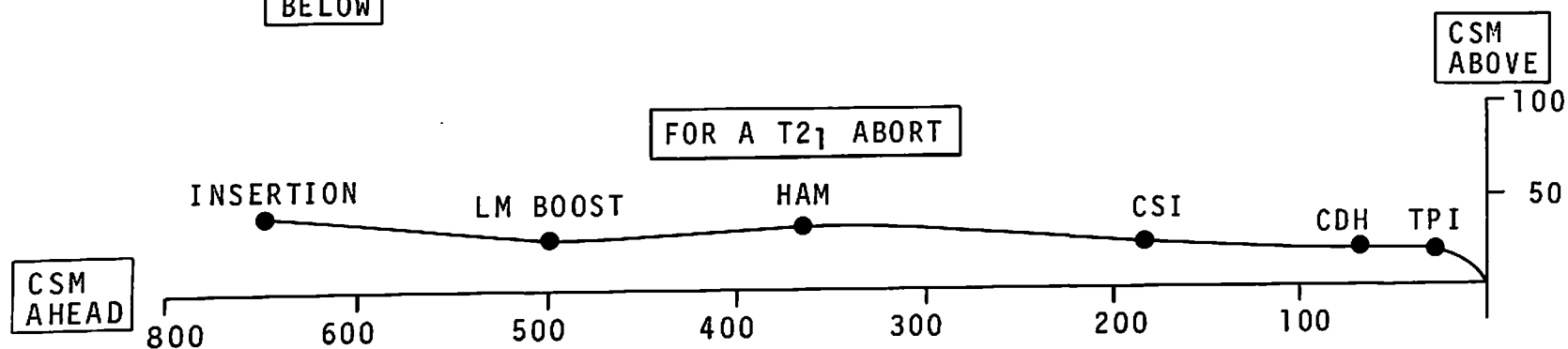
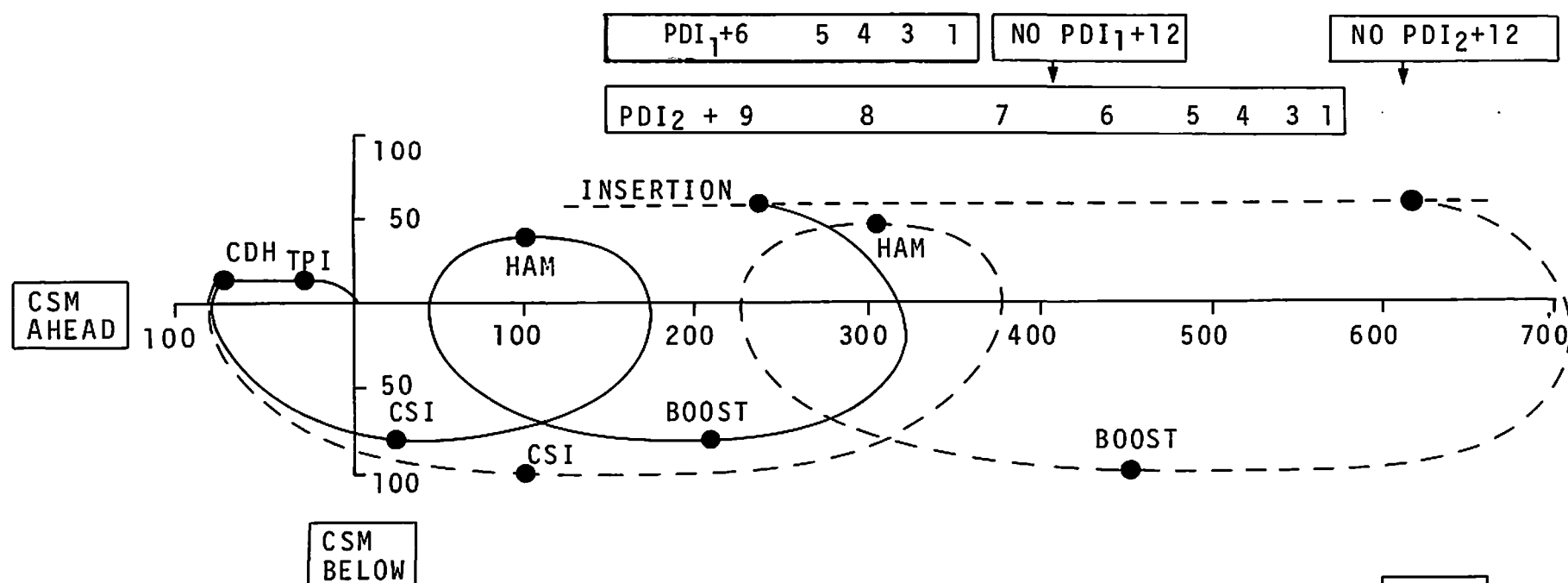
MCC COMPUTATION

		MCC2 DV-LV		
FIN COMP	81			
P76	84			



TWO REV
RELATIVE MOTION PLOT

LM CENTERED



**TWO REV
BURN TIMES**

NO PDI 1+12 ABORT

BOOST AT AIM +55 MIN
HAM AT BOOST +60 MIN
CSI AT HAM +60 MIN
TPI AT 109:18:38.3

PDI 1, EARLY ABORT

BOOST AT INS +60 MIN
HAM AT BOOST +60 MIN
CSI AT HAM +60 MIN
TPI AT 109:18:38.3

T2 ABORT, 1ST OPP

BOOST AT INS +50 MIN
HAM AT BOOST +60 MIN
CSI AT HAM +50 MIN
TPI AT 109:18:38.3

NO PDI 2+12 ABORT

BOOST AT AIM +60 MIN
HAM AT BOOST +60 MIN
CSI AT HAM +60 MIN
TPI AT 111:17:04.2

PDI 2, EARLY ABORT

BOOST AT INS +60 MIN
HAM AT BOOST +60 MIN
CSI AT HAM +60 MIN
TPI AT 111:17:04.2

TWO REV TIMELINE

POWER DOWN SIM(SEE SYSTEMS CHECKLIST, PAGE S/1-40)
POO; V48(11102,X1111)

INSERTION OR A. I. M.
MSFN UPLINK: LM S.V.

OBTAIN LM BOOST GETI AND DV(COPY)

LM BOOST

P76; OBTAIN LM CSI, TPI AND HAM GETI'S(COPY)
P31 WITH LM GETI(SEE INPUT)

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

RECYCLE(COPY)

FINAL COMP, OVERWRITE DVY (COPY);
OBTAIN LM HAM ΔV(COPY)
P41m

HAM

P76m

IF NO MARKS HAVE BEEN TAKEN:
V67(+10000,+00100,+00001)

INSERTION		LM INSERTION TIME		
A.I.M. P76	33	LM AIM GETI	:	:
	84	LM AIM ΔV-LV	:	:

LM BOOST P76	33	LM BOOST GETI	:	:
	84	LM BOOST ΔV-LV	+0000.0	+0000.0

HAM COMPUTATION				
INPUT	11	LM CSI GETI	:	:
	55	H +00001	ELEVATION +208.30	180° OPT +130.00
	37	LM TPI GETI	:	:
	33	LM HAM GETI	:	:
RECYCLE	90	Y	Y DOT	LM Y DOT
	81	HAM ΔV-LV	:	+0000.0
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	HAM ΔV-LV	+0000.0	+0000.0
P76	84	LM HAM ΔV-LV	:	+0000.0

DATE 6/22/77

PAGE 11

P32m

(10000/10)W-MATRIX

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

(2000/2)W-MATRIX

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

RECYCLE(COPY)

FINAL COMP(COPY); OBTAIN LM CSI DV(COPY); P40m

CSI BURN

P76m; OBTAIN LM PC GETI(COPY); P36m WITH LM GETI

FINAL COMP (COPY) TO BYPASS P52m AND P4Xm, ZERO DVY
OBTAIN LM PC DV(COPY)
P52m;P4Xm

PLANE CHANGE

TO BYPASS GYRO TORQUE: ENTER ON F 50 25
TO GYRO TORQUE: CMC MODE-FREE
PRO ON F 50 25

P76m; OBTAIN LM CDH GETI (COPY); P33m WITH LM GETI

RECYCLE (COPY)

FINAL COMP (COPY); OBTAIN LM CDH DV (COPY); P40m

CDH BURN

P76m

CSI COMPUTATION

INPUT	11	LM CSI GETI		PROCEED ON DISPLAYED GETI
	55	N	ELEVATION	180° OPT
	37	+00001	+208.30	+130.00
RECYCLE	90	LM TPT GETI	Y DOT	LM Y DOT
	81	Y	CSI ΔV-LV	+0000.0
	81	Y	CSI ΔV-LV	+0000.0
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	Y	Y DOT	LM Y DOT
	81	Y	Y DOT	LM Y DOT
P76	84	LM CSI ΔV-LV	Y DOT	LM Y DOT
	84	Y	Y DOT	LM Y DOT
	84	Y	Y DOT	LM Y DOT

PLANE CHANGE COMPUTATION

INPUT	33	LM PC GETI		PROCEED ON DISPLAYED GETI
	90	Y	Y DOT	LM Y DOT
	81	Y	PC ΔV-LV	+0000.0
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	Y	PC ΔV-LV	+0000.0
	81	Y	PC ΔV-LV	+0000.0
P76	84	LM PC ΔV-LV	Y DOT	LM Y DOT
	84	Y	Y DOT	LM Y DOT
	84	Y	Y DOT	LM Y DOT

CDH COMPUTATION

INPUT	13	LM CDH GETI		PROCEED ON DISPLAYED GETI
	90	Y	Y DOT	LM Y DOT
	81	Y	CDH ΔV-LV	+0000.0
RECYCLE	90	Y	Y DOT	LM Y DOT
	81	Y	CDH ΔV-LV	+0000.0
	81	Y	CDH ΔV-LV	+0000.0
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	Y	Y DOT	LM Y DOT
	81	Y	Y DOT	LM Y DOT
P76	84	LM CDH ΔV-LV	Y DOT	LM Y DOT
	84	Y	Y DOT	LM Y DOT
	84	Y	Y DOT	LM Y DOT

P34m, OVERWRITE N55 R2

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

IF ONLY ONE SENSOR OPERATIONAL:
V57(.00004, .00001)

RECYCLE(COPY)

OBTAIN LM TPI GETI AND DV(COPY); FINAL
COMP, OVERWRITE N37 WITH LM TPI GETI

P40m

TPI BURN

P76m; P35m

FINAL COMP(COPY);OBTAIN LM MCC1 DV(COPY); P41m

MCC1

P76m; P35m

FINAL COMP(COPY); OBTAIN LM MCC2 DV(COPY); P41m

MCC2

GO TO TPF ACTIVITIES(PAGE 38)

TPF

TPI-ELEVATION OPTION

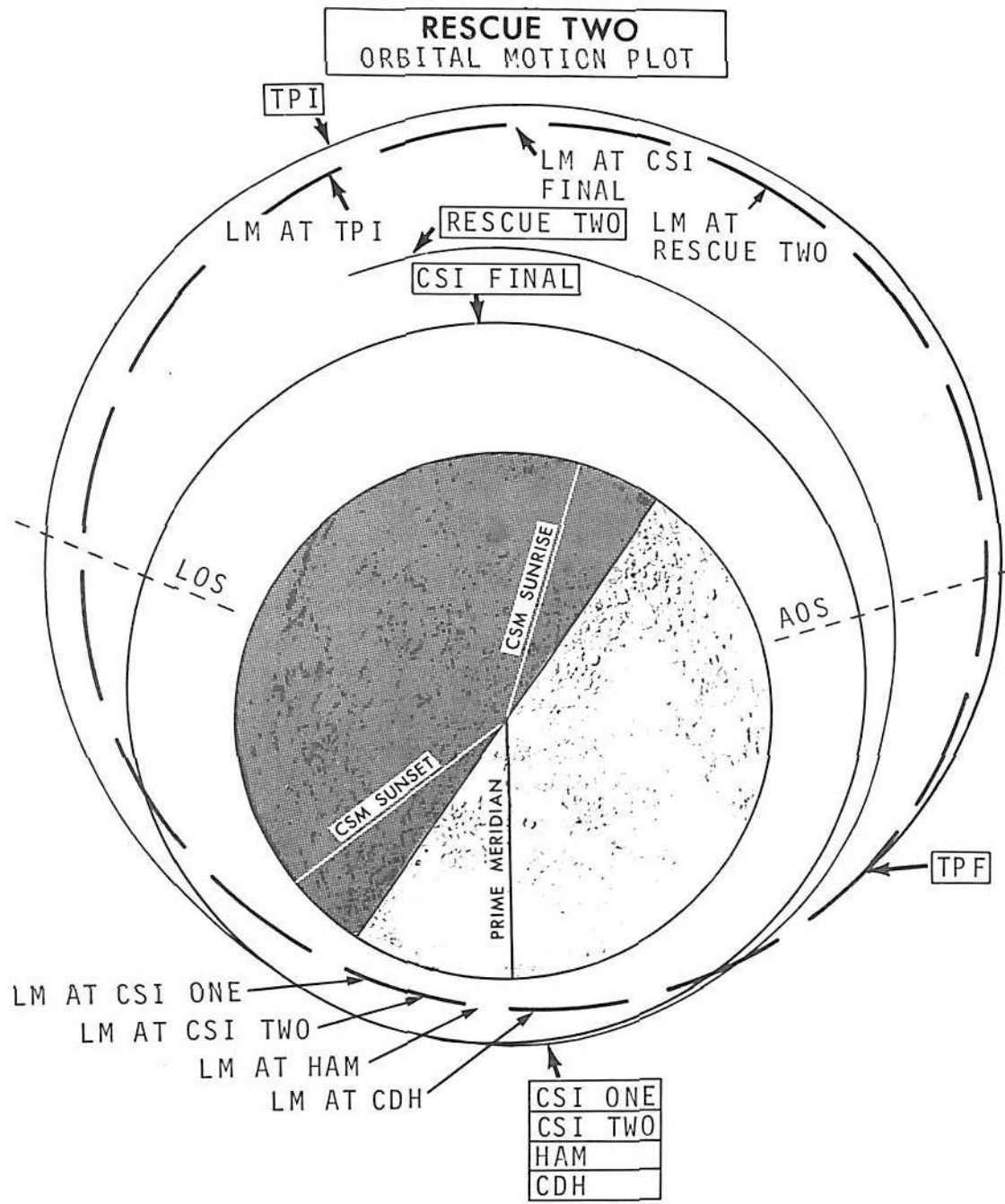
INPUT		TPI GETI		
		INTEG OPT	ELEVATION *	TRANSFER *
	37			
	55	+00000	+208.30	+130.00
RECYCLE	37			
	58			
	81			
FINAL COMP	37			
	55	+00000		+130.00
	58			
P76	33			
	84			

MCC COMPUTATION

FIN COMP		MCC1 DV-LV		
	81			
P76	84			

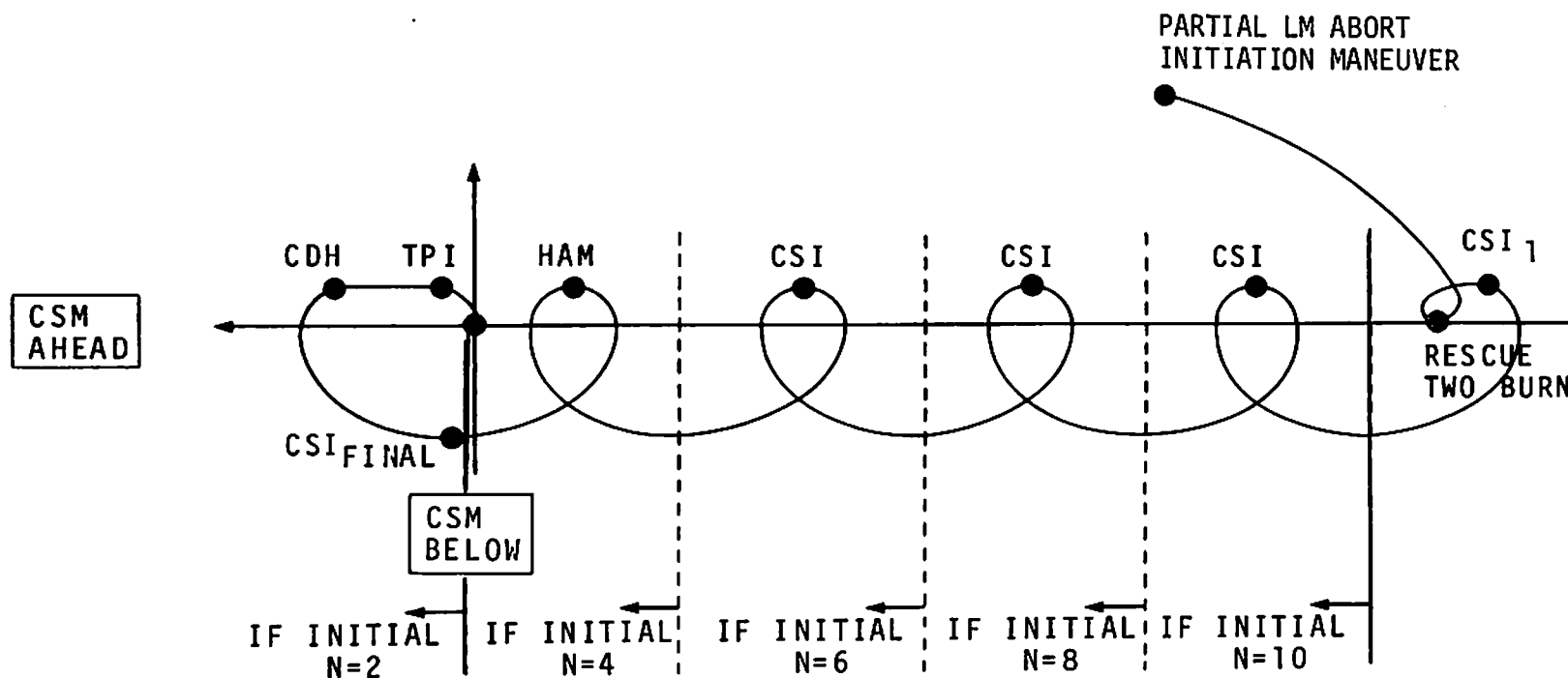
MCC COMPUTATION

FIN COMP		MCC2 DV-LV		
	81			
P76	84			



RESCUE TWO RELATIVE MOTION PLOT

LM CENTERED



RESCUE TWO
TIMELINE

POWER DOWN SIM(SEE SYSTEM CHECKLIST,
PAGE S/1-40); P00; V48(11102,X111)

INSERTION OR A.I.M.

MSFN UPLINK: LM S.V.
MSFN UPDATE: RESCUE TWO P30 PAD
P32 INITIALIZATION PAD

P30; P40

RESCUE TWO BURN

P00

RESCUE TWO P30 PAD

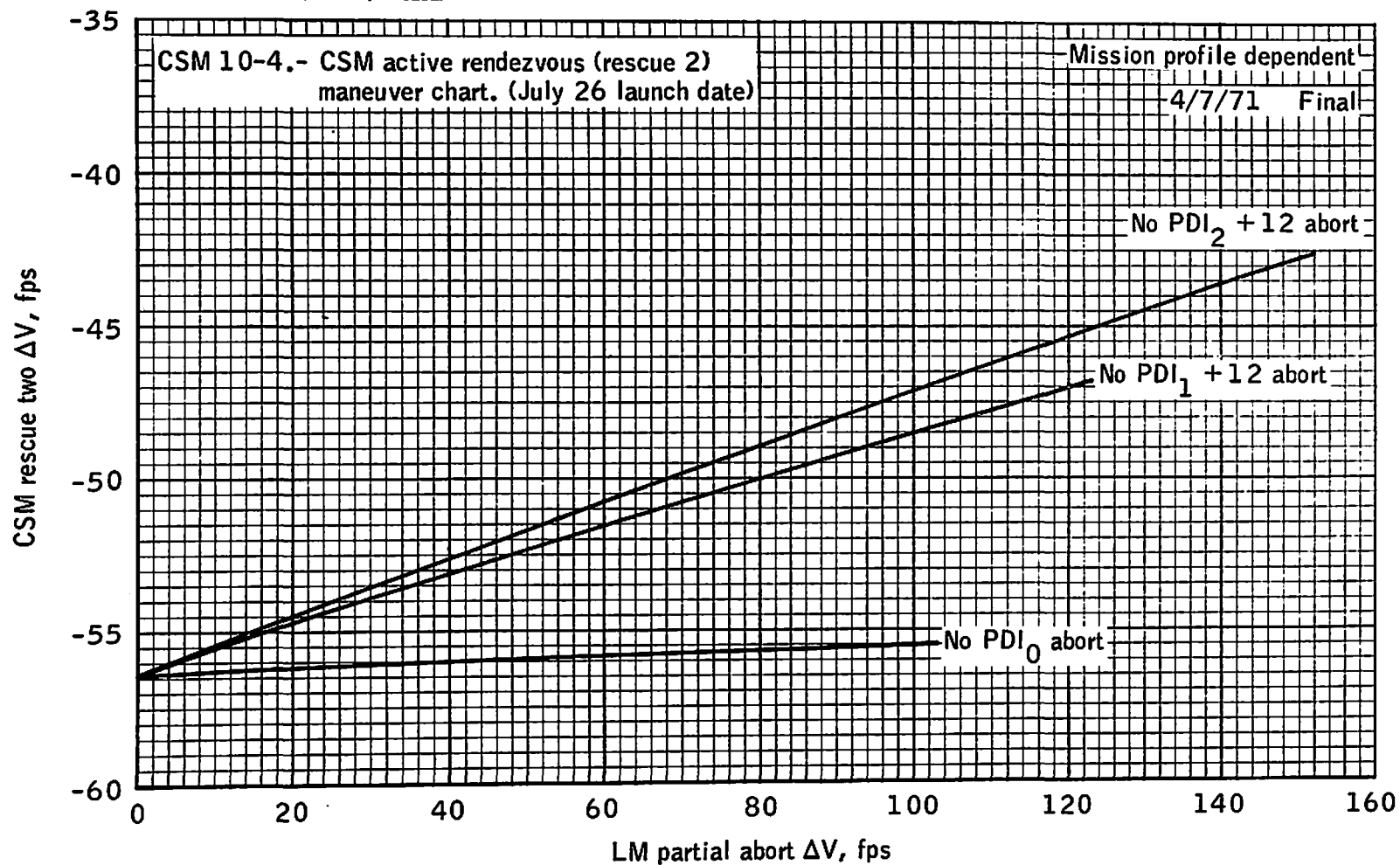
SET STARS	R	2	N	=			PURPOSE
	S	P	S	G	E	N	PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
	+	0	0	0	0	0	ΔV_Y
	+	0	0	0	0	0	ΔV_Z
	X	X	X				R
	X	X	X				P
	X	X	X				Y
OTHER	+						H _A N44
							H _P
	+						ΔVT
	X	X	X				BT
	X						ΔVC

P32 INITIALIZATION PAD

N11	CSI GETI	+		:	:	:	.
	N	+	0	0	0		
N55	ELEV ∇	+	2	0	8	3	0
	180° OPT	+	1	3	0	0	0
N37	TPI GETI	+		:	:	:	.

RESCUE TWO BURN INFORMATION

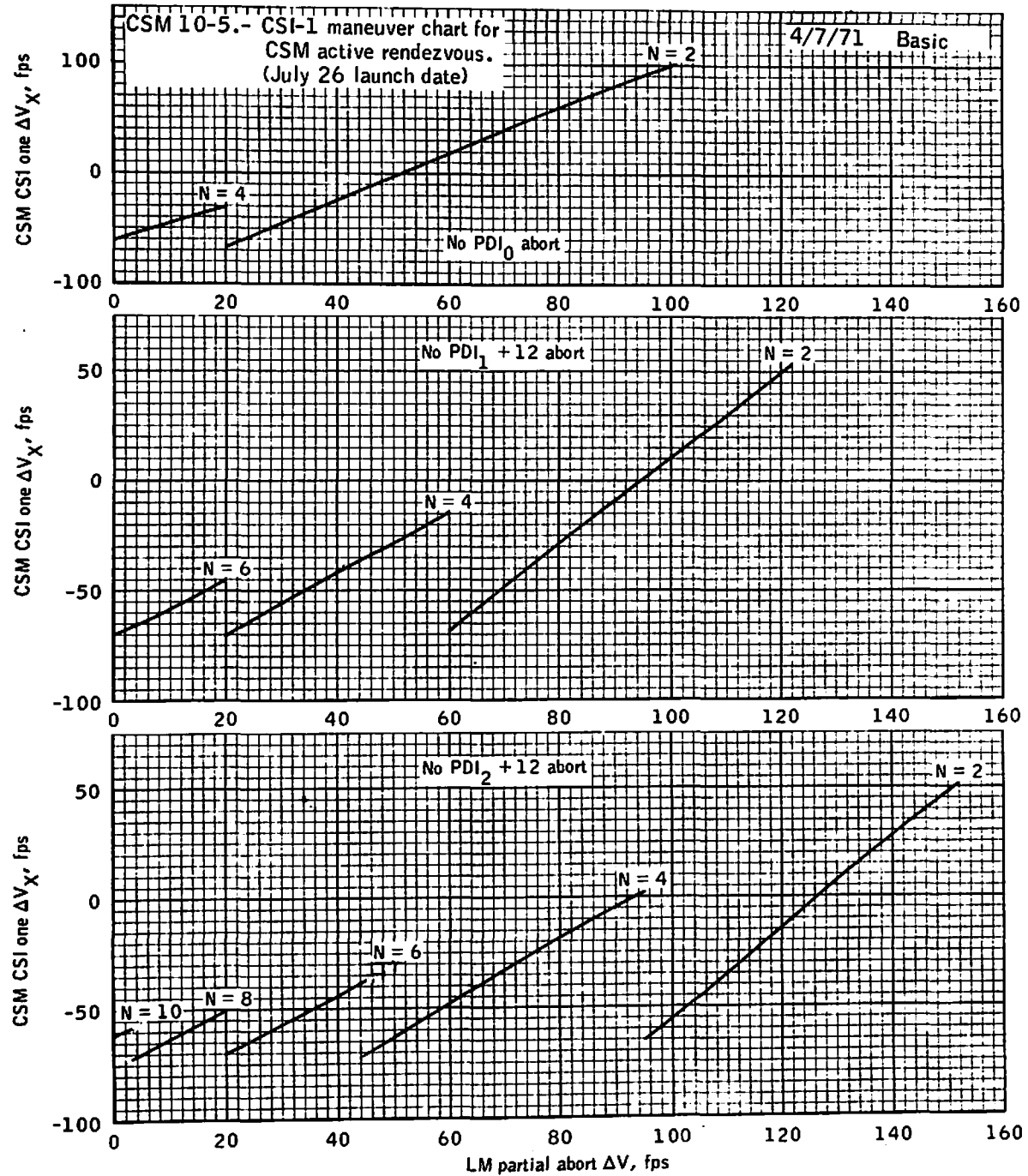
DuPont/OMAB/MPAD (for CSM Rescue)

Data source Spacecraft O.T.Data confirmed ASD

CSM active rendezvous (rescue 2) maneuver chart.

CSI ONE BURN INFORMATION

DuPont/OMAB/MPAD (for CSM Rescue)
 Data source SC O.T.
 Data confirmed U.S.



P32 (LOAD P32 INITIALIZATION PAD ON PAGE 16)

BEFORE STEADY STATE
N49(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49(+00030,+00020):REJECT/REPEAT

V
H
F
O
N
L
Y

-12

0

FINAL COMP, OVERWRITE DVY(COPY); P40m

CSI ONE BURN

P76m

IF NO MARKS HAVE BEEN TAKEN:
V67(+10000,+00100,+00001)

IF INITIAL N>4, CONTINUE ON PAGE 20
IF INITIAL N=4, CONTINUE ON PAGE 21
IF INITIAL N=2, CONTINUE ON PAGE 22

P32	90	Y	Y DOT	LM Y DOT
FINAL		.	.	.
COMP	81	CSI 3V-LV	+0000.0	+0000.0

GROUND
 $\Delta V X$

.

COMPARISON
LIMIT=±3 FPS

(A)

P32m, **REDUCE PREVIOUS N BY 2**, OVERWRITE N55 R1(COPY)

P52,OPT 3; P32

(10000/10)W-MATRIX

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

(2000/2)W-MATRIX

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

IF NECESSARY: MSFN UPLINK-LM S.V.
CSM S.V.

FINAL COMP, OVERWRITE DVY(COPY);

P41m

CSI BURN

P76m

IF NO MARKS HAVE BEEN TAKEN:
V67(+10000,+00100,+00001)

RECYCLE TO (A) UNTIL
N=4 WAS INPUT FOR THE
PREVIOUS CSI BURN,
THEN CONTINUE TO PAGE 21.

FOR N=8

CSI ☐ COMPUTATION			
INPUT	11	CSI GETI :	

P31m (COPY), LOAD $\Delta H = 10.0$

V24	N1E
2367E	
0E	
22062E	

(10000/10)W-MATRIX

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

(2000/2)W-MATRIX

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

HAM COMPUTATION			
INPUT	11	CSI GETI	
		PROCEED ON DISPLAYED GETI	
	55	N +00001	ELEVATION 1 +208.30
	37	TPI GETI	REC' OPT +130.00
RECYCLE	33	HAM GETI	
		PROCEED ON DISPLAYED GETI	
	90	Y	Y DOT
	81	HAM AV-LV	LP Y DOT
FINAL COMP			+0000.0
	90	Y	Y DOT
			LP Y DOT
	81	HAM JV-LV	
			+0000.0
			+0000.0

RECYCLE (COPY)

FINAL COMP, OVERWRITE DVY(COPY); P41m

HAM

P76m

IF NO MARKS HAVE BEEN TAKEN: V67(+10000,+00100,+00001)

P32m(COPY)

IF P36 IN MODE LITE:
V37E 32E

(10000/10) W-MATRIX)

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

(2000/2) W-MATRIX

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

SXT/VHF

RECYCLE(COPY)

FINAL COMP(COPY); P41m

CSI FINAL BURN

P76m; P36m (COPY)

SXT/VHF

FINAL COMP (COPY) TO BYPASS P52m AND P4Xm, ZERO DVY
P52m; P4Xm

PLANE CHANGE

P76m; P33m(COPY)

SXT/VHF

RECYCLE(COPY)

FINAL COMP(COPY); P40m

CDH BURN

P76m

TO BYPASS GYRO TORQUE: ENTER ON F 50 25
TO GYRO TORQUE: CMC MODE-FREE
PRO ON F 50 25

CSI COMPUTATION

INPUT	11	CSI GETI		
	55	PROCEED ON DISPLAYED GETI		
	37	N	ELEVATION	160 OPT
		+00001	+208.30	+130.00
RECYCLE	90	TPI GETI		
	81	PROCEED ON DISPLAYED GETI		
		Y	Y DOT	LM Y DOT
		CSI :V-LV	.	+0000.0
FINAL COMP	90			
	81	Y	Y DOT	LM Y DOT
		CSI :V-LV	.	+0000.0

PLANE CHANGE COMPUTATION

INPUT	33	PC GETI		
		PROCEED ON DISPLAYED GETI		
		Y	Y DOT	LM Y DOT
FINAL COMP	90			
	81	PC :V-LV	.	+0000.0
			.	

CDH COMPUTATION

INPUT	13	LM CDH GETI		
		PROCEED ON DISPLAYED GETI		
		Y	Y DOT	LM Y DOT
RECYCLE	90			
	81	CDH :V-LV	.	
			.	
FINAL COMP	90			
	81	Y	Y DOT	LM Y DOT
		CDH :V-LV	.	

-20 SXT/VHF P34m, OVERWRITE N55 R2
 RECYCLE (COPY)

-12 FINAL COMP (COPY); P40m

0 TPI BURN
 P76m; P35m

+12 S/V P35 FINAL COMP (COPY); P41m
 +15 MCC1
 P76m, P35m

+27 S/V P35 FINAL COMP (COPY); P41m
 +30 MCC2
 GO TO TPF ACTIVITIES (PAGE 38)

+42 TPF

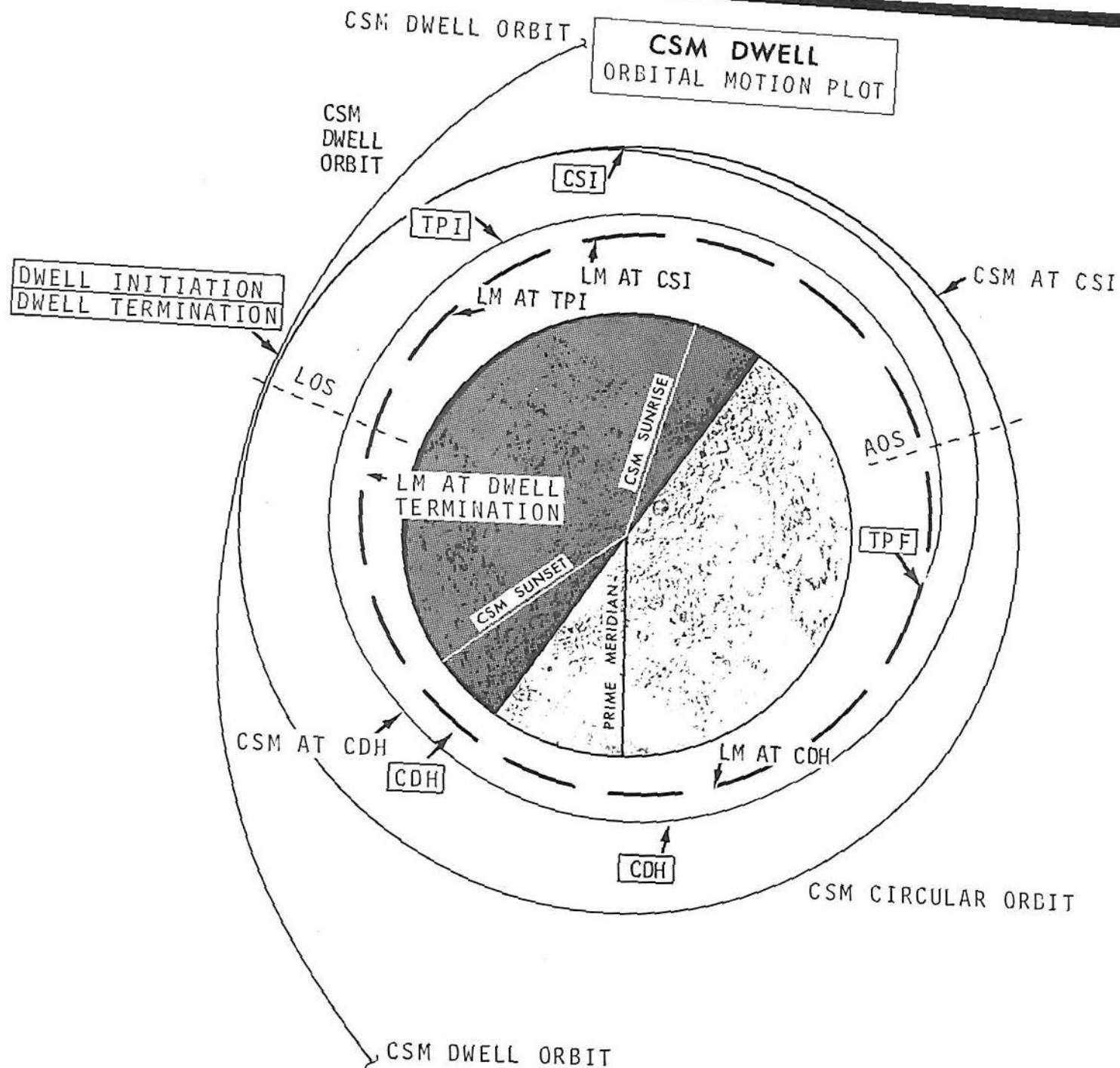
BEFORE STEADY STATE
 N49>(+00080,+00050):REJECT/REPEAT
 AFTER STEADY STATE
 N49>(+00030,+00020):REJECT/REPEAT

IF ONLY ONE SENSOR OPERATIONAL:
 V57(00004, 00001)

TPI-ELEVATION OPTION			
INPUT	37	TPI GETI	
	55	PROCEED ON DISPLAYED GETI	
		INTLG OPT	ELEVATION
		+00000	+208.30
			+130.00
RECYCLE	37	TPI GETI	
	58	PROCEED ON DISPLAYED GETI	
	81		
		PERILUNE ALT	TPF 2V
		TPI 2V-LV	
FINAL COMP	37	TPI GETI	
	55	PROCEED ON DISPLAYED GETI	
	81		
		INTLG OPT	ELEVATION
		+00000	+130.00
		PERILUNE ALT	TPF 2V
		TPI 2V-LV	

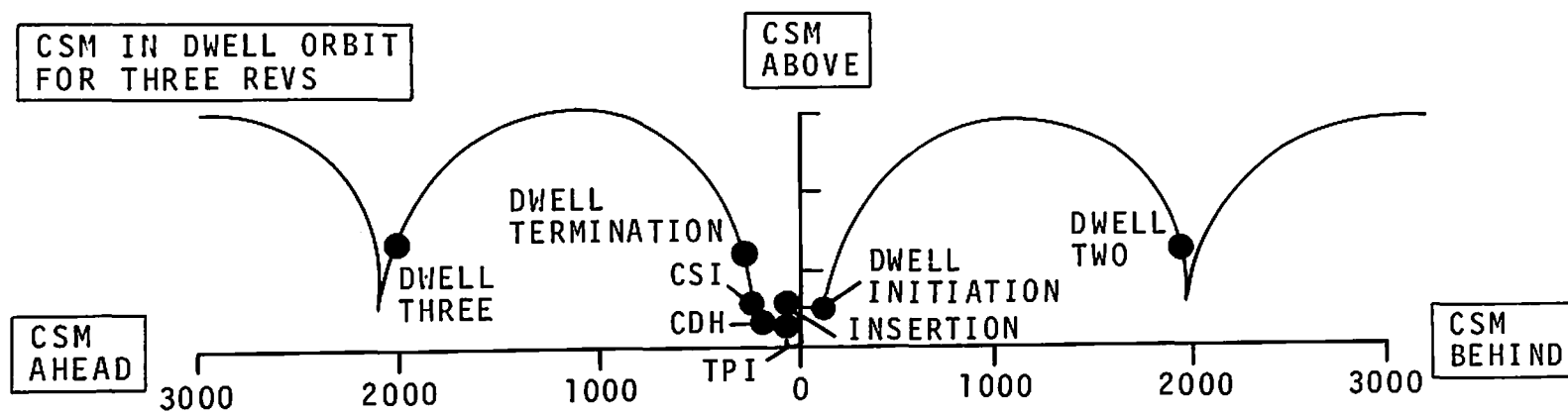
MCC COMPUTATION			
FIN COMP	81	MCC1 2V-LV	

MCC COMPUTATION			
FIN COMP	81	MCC2 2V-LV	



CSM DWELL RELATIVE MOTION PLOT

LM CENTERED



CSM DWELL
TIMELINE

DWELL INITIATION P30 PAD

	D	W	E	L	L	I	PURPOSE
SET STARS	S	P	S/G	&	N		PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
	X	X	X				Y
OTHER	+						H _A N44
REMAIN IN							H _P
THE DWELL							
ORBIT FOR	+						ΔVT
REVS	X	X	X				BT
	X						ΔVC

P32 INITIALIZATION PAD

N11	CSI GETI	+	:	:	:	:
	N	+	0	0	0	1
N55	ELEV &	+	2	0	8	3
	180° OPT	+	1	3	0	0
N37	TPI GETI	+	:	:	:	:

POWER DOWN SIM(SEE SYSTEMS CHECKLIST, PAGE S/1-40)

INSERTION

MSFN UPLINK: LM S.V.

MSFN UPDATE: DWELL INITIATION P30 PAD (COPY)

P32 INITIALIZATION PAD (COPY)

P30; P40

DWELL INITIATION BURN

P00

DWELL TWO P30 PAD

	D	W	E	L	L	2	PURPOSE
SET STARS				G	&	N	PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
	X	X	X				Y
	+						H _A N44
							H _P
	+						ΔVT
	X	X	X				BT
	X						ΔVC

MSFN UPDATE: DWELL TWO P30 PAD (COPY)
P32 INITIALIZATION PAD (COPY)

P52, OPT 3
P00

P30; P41

DWELL TWO BURN

P00

P32 INITIALIZATION PAD

N11	CSI GETI	+	: : .					
N55	N	+	0	0	0	0	1	
	ELEV 4	+	2	0	8	3	0	
	180° OPT	+	1	3	0	0	0	
N37	TPI GETI	+	: : .					

-9

0

THIS PAGE IS FOR A
THREE REV DWELL
ONLY.

IF DWELL TERMINATION
IS THE NEXT BURN,
GO TO PAGE 29.

MSFN UPDATE: DWELL THREE P30 PAD(COPY)
P32 INITIALIZATION PAD(COPY)

P52, OPT 3
P00

P30; P41

DWELL THREE BURN

P00

DWELL THREE P30 PAD

	D	W	E	L	L	3	PURPOSE
SET STARS				G	&	N	PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
	X	X	X				Y
	+						H _A N44
							H _P
	+						ΔVT
	X	X	X				BT
	X						ΔVC

P32 INITIALIZATION PAD

N11	CSI GETI	+	: : .					
N55	N	+	0	0	0	0	1	
	ELEV	+	2	0	8	3	0	
	180° OPT	+	1	3	0	0	0	
N37	TPI GETI	+	: : .					

DWELL TERMINATION PAD

SET STARS	D	W	E	L	L	T	PURPOSE
	S	P	S	G	&	N	PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
OTHER	X	X	X				Y
BURN INCLUDES	+						H _A N44
PLANE CHANGE							H _P
COMPONENT	+						ΔVT
	X	X	X				BT
	X						ΔVC

P32 INITIALIZATION PAD

N11	CSI GETI	+	:	:	:	:
	N	+	0	0	0	0 1
N55	ELEV &	+	2	0	8	3 0
	180° OPT	+	1	3	0	0 0
N37	TPI GETI	+	:	:	:	:

MSFN UPDATE: DWELL TERMINATION
P30 PAD INCLUDING
OUT OF PLANE COMPONENT (COPY)
P32 INITIALIZATION PAD (COPY)

P52, OPT 3
P00
V48(11102,X1111)
P30; P40

DWELL TERMINATION BURN

P32 (LOAD P32 INITIALIZATION PAD)

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

FINAL COMP(COPY); P40m

CSI BURN

P76m; P36m(COPY)

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

FINAL COMP(COPY), ZERO DVY TO BYPASS P52m,P4Xm
P52m; P4Xm

PLANE CHANGE

P76m; P33m(COPY)

TO BYPASS GYRO TORQUE: ENTER ON F 50 25
TO GYRO TORQUE: CMC MODE-FREE
PRO ON F 50 25

RECYCLE(COPY)

FINAL COMP(COPY); P40m

CDH BURN

P76m

CSI COMPUTATION

FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	CSI ΔV-LV	.	+0000.0

PLANE CHANGE COMPUTATION

INPUT	33	PC GETI	PROCEED ON DISPLAYED GETI
		Y	Y DOT
FINAL COMP	90	Y	LM Y DOT
	81	PC ΔV-LV	+0000.0

CDH COMPUTATION

INPUT	13	CDH GETI	PROCEED ON DISPLAYED GETI
		Y	Y DOT
RECYCLE	90	Y	LM Y DOT
	81	CDH ΔV-LV	.
FINAL COMP	90	Y	Y DOT
	81	CDH ΔV-LV	.

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

IF ONLY ONE SENSOR OPERATIONAL:
V57(00004, 00001)

P34m, OVERWRITE N55 R2

-20 SXT/VHF RECYCLE(COPY)

-12 FINAL COMP(COPY); P40m

0 TPI BURN
P76m; P35m

+12 S/V FINAL COMP (COPY); P41m

+15 MCC1
P76m, P35m

+27 S/V FINAL COMP(COPY); P41m

+30 MCC2
GO TO TPF ACTIVITIES (PAGE 38)

+42 TPF

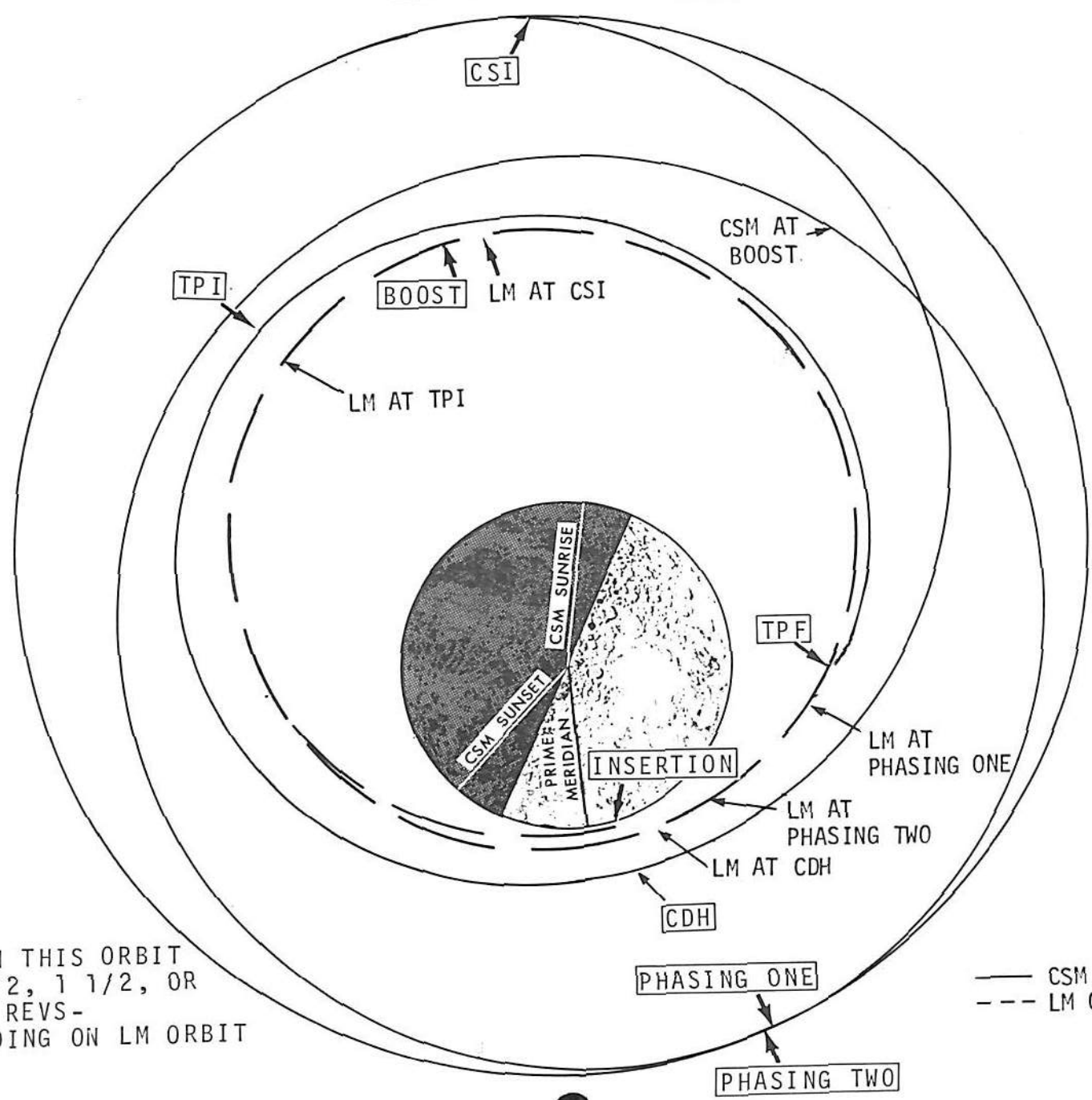
TPI ELEVATION OPTION			
INPUT	37	TPI GETI	
	55	PROCEED ON DISPLAYED GETI	
		INTEG OPT	ELEVATION
		+00000	+208.30
			TRANSFER
			+130.00
RECYCLE	37	TPI GETI	
		PROCEED ON DISPLAYED GETI	
	58	PERILUNE ALT	TPI ΔV
	59	TPI ΔV-LOS	TPF ΔV
FINAL COMP	37	TPI GETI	
		PROCEED ON DISPLAYED GETI	
	58	PERILUNE ALT	TPI ΔV
	81	TPI ΔV-LV	TPF ΔV
	59	TPI ΔV-LOS	TPF ΔV

MCC COMPUTATION			
FIN COMP	81	MCC1 ΔV-LV	

MCC COMPUTATION			
FIN COMP	81	MCC2 ΔV-LV	

MANUAL INSERTION ORBITAL MOTION PLOT

CSM IN THIS ORBIT
FOR 1/2, 1 1/2, OR
2 1/2 REVS-
DEPENDING ON LM ORBIT

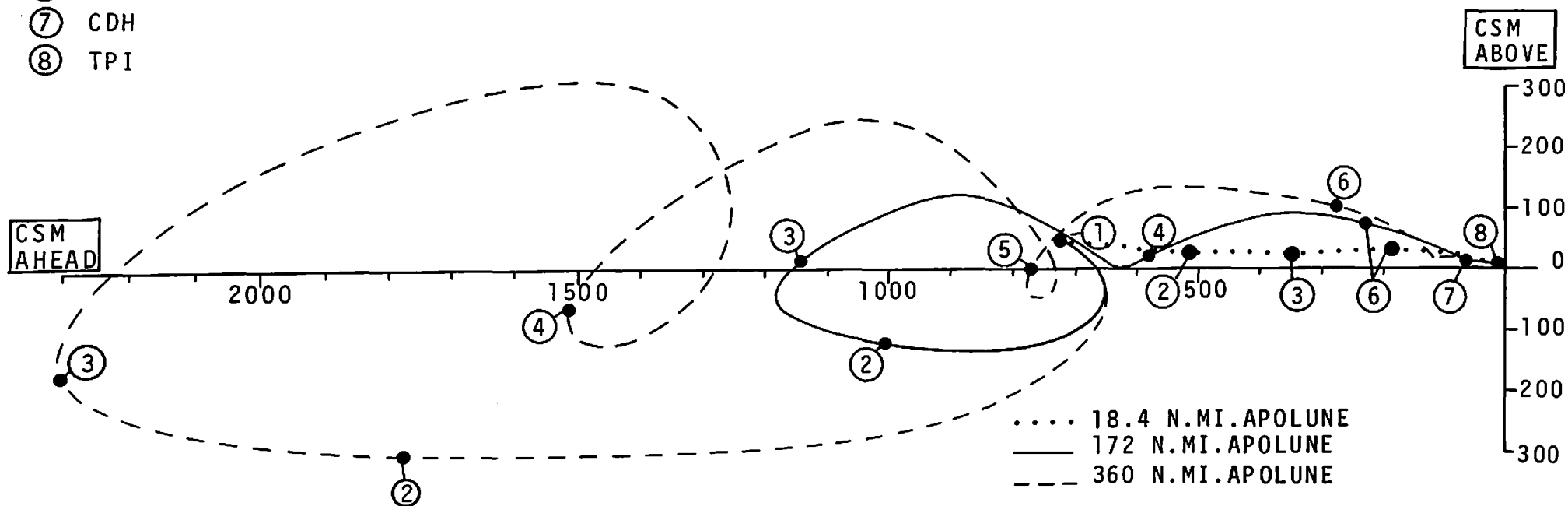


— CSM ORBITS
--- LM ORBITS

MANUAL INSERTION RELATIVE MOTION PLOTS

LM CENTERED

- ① INSERTION
- ② LM BOOST
- ③ PHASING ONE
- ④ PHASING TWO (NOMINALLY ZERO)
- ⑤ PHASING THREE (NOMINALLY ZERO)
- ⑥ CSI
- ⑦ CDH
- ⑧ TPI



MANUAL INSERTION
TIMELINE

POWER DOWN SIM(SEE SYSTEMS CHECKLIST, PAGE S/1-40)

INSERTION

MSFN UPLINK: LM S.V.

P52, OPT 3
P00

LM BOOST P76	33 84	LM GETI-BOOST LM BOOST ΔV -LV +0010.0	:	:	:	:
		+0000.0		+0000.0		+0000.0

LM BOOST

VERIFY LM BOOST BURN
P76

MSFN UPDATE: PHASING P30 PAD (COPY)
P32 INITIALIZATION PAD (COPY)

P30; P40

PHASING ONE

PHASING P30 PAD

	P	H	A	S	E	1	PURPOSE
SET STARS	S	P	S	G	&	N	PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
OTHER	X	X	X				Y
	+						H _A N44
							H _P
	+						ΔVT
	X	X	X				BT
	X						ΔVC

P32 INITIALIZATION PAD

N11	CSI GETI	+	:	:	:	:
	N	+	0	0	0	0 1
N55	ELEV & 180° OPT	+	2	0	8	3 0
		+	1	3	0	0 0
N37	TPI GETI	+	:	:	:	:

PHASING TWO P30 PAD

	P	H	A	S	E	2	PURPOSE
SET STARS			/G	&	N		PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
	X	X	X				Y
	+						H _A N44
							H _P
	+						ΔVT
	X	X	X				BT
	X						ΔVC

P32 INITIALIZATION PAD

N11	CSI GETI	+	:	:	.
	N	+	0	0	0 0 1
N55	ELEV 4	+	2	0	8.3 0
	180° OPT	+	1	3	0.0 0
N37	TPI GETI	+	:	:	.

PHASING THREE P30 PAD

	P	H	A	S	E	3	PURPOSE
SET STARS			/G	&	N		PROP/GUID
	+						WT N47
R ALIGN		0	0				P TRIM N48
P ALIGN		0	0				Y TRIM
Y ALIGN	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
	X	X	X				Y
	+						H _A N44
							H _P
	+						ΔVT
	X	X	X				BT
	X						ΔVC

P32 INITIALIZATION PAD

N11	CSI GETI	+	:	:	.
	N	+	0	0	0 0 1
N55	ELEV 4	+	2	0	8.3 0
	180° OPT	+	1	3	0.0 0
N37	TPI GETI	+	:	:	.

P52, OPT 3
P00MSFN UPDATE:
PHASING P30 PAD
(COPY)
P32 INITIALIZATION
PAD (COPY)

P30; P41

PHASING

P00

IF THE NEXT BURN IS NOT CSI,
RETURN TO (A)

V48(11102,X1111)
P32(LOAD P32 INITIALIZATION PAD)

BEFORE STEADY STATE
N49>(+00200,+00120):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

CSI COMPUTATION				
RECYCLE	90	Y	Y DOT	LM Y DOT
	81	CSI ΔV-LV	.	.
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	CSI ΔV-LV	.	+0000.0

-20

RECYCLE(COPY)

-12

FINAL COMP(COPY); P40m

0

CSI BURN

P76m; P36m(COPY)

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

-9

FINAL COMP(COPY), ZERO DVY TO BYPASS P52m,P4Xm
P52m; P4Xm

0

PLANE CHANGE

P76m; P33m(COPY)

TO BYPASS GYRO TORQUE: ENTER ON F 50 25
TO GYRO TORQUE: CMC MODE-FREE
PRO ON F 50 25

-20

RECYCLE(COPY)

-12

FINAL COMP(COPY); P40m

0

CDH BURN

P76m

PLANE CHANGE COMPUTATION				
INPUT	33	PC GETI	PROCEED ON DISPLAYED GETI	
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	PC ΔV-LV	.	+0000.0

CDH COMPUTATION				
INPUT	13	CDH GETI	PROCEED ON DISPLAYED GETI	
RECYCLE	90	Y	Y DOT	LM Y DOT
	81	CDH ΔV-LV	.	.
FINAL COMP	90	Y	Y DOT	LM Y DOT
	81	CDH ΔV-LV	.	.

-20 SXT/VHF
P34m, OVERWRITE N55 R2
RECYCLE (COPY)

-12
FINAL COMP (COPY); P40m

0
TPI BURN
P76m; P35m
S/V
FINAL COMP(COPY); P41m
+12
MCC1
P76m; P35m
S/V
FINAL COMP(COPY); P41m
+27
MCC2
GO TO TPF ACTIVITIES (PAGE 38)
+30
+42
TPF

BEFORE STEADY STATE
N49>(+00080,+00050):REJECT/REPEAT
AFTER STEADY STATE
N49>(+00030,+00020):REJECT/REPEAT

IF ONLY ONE SENSOR OPERATIONAL:
V57(00004, 00001)

TPI-ELEVATION OPTION				
INPUT	37	TPI GETI PROCEED ON DISPLAYED GETI		
	55	INTEG OPT +00000	ELEVATION +208.30	TRANSFER +130.00
	81	PERILUNE ALT	TPI ΔV	TPF ΔV
RECYCLE	37	TPI GETI PROCEED ON DISPLAYED GETI		
	58	PERILUNE ALT	TPI ΔV	TPF ΔV
	81	TPI ΔV-LV		
FINAL COMP	37	TPI GETI PROCEED ON DISPLAYED GETI		
	55	INTEG OPT +00000	ELEVATION .	TRANSFER +130.00
	81	PERILUNE ALT	TPI ΔV	TPF ΔV

MCC COMPUTATION			
FIN COMP	81	MCC1 ΔV-LV	

MCC COMPUTATION			
FIN COMP	81	MCC2 ΔV-LV	

TPF ACTIVITIES

+30

(11102)
(X1111)

MCC2

P76m; P79m

ACQ MSFN

PERFORM PRE-DOCK CHECKLIST

IF CSM ACTIVE
P47 AT 1.25 N.M.
SEC PRPLNT FUEL PRESS(4)-OPEN
V83E ←
N83E
KEY REL

UTILITY PWR-ON (VERIFY)

TV-ON

DAC-ON

LM PHOTOS WITH DAC/TV

TPF

EMS MODE-STBY

EMS FUNC-OFF

EXT RNDZ LT-OFF

LM STATION KEEP

P00

DAC/TV-OFF

+42

PRE-DOCK CHECKLIST

MAN ATT(3)-RATE CMD(VERIFY)
LIMIT CYCLE-OFF(VERIFY)
ATT DB-MIN
RATE-LOW(VERIFY)
TRANS CONTR PWR-ON(UP)
ROT CONTR PWR DIRECT(BOTH)-MNA/MNB
SC CONT-CMC(VERIFY)
CMC MODE-HOLD
AUTO RCS SEL(16)-MNA/MNB(VERIFY)
CB DOCK PROBE(2)-CLOSED
PROBE RETR(2)-OFF(VERIFY)
PROBE EXTD/REL-RETR
PROBE EXTD/REL TB(2)-GRAY(VERIFY)
(IF TB NOT GRAY, GO TO PG S/2-12,E)
CB SECS LOGIC(2)-CLOSED(VERIFY)
CB SECS ARM(2)-CLOSED
EXT LIGHTS RUN/EVA-ON(UP)(VERIFY)
COAS PWR-ON(UP)(VERIFY)

BRAKING GATES

R,NN	Ṙ,FPS	RETICLE ANG,DEG	R,FT
1.50	45	.08	9000
1.00	30	.13	6000
.50	20	.26	3000
.25	10	.54	1500
.08	5	1.60	500
.05		2.70	300
.03		4.00	200
.02		8.50	100

CMC MODE - AUTO
V49 MNVR TO SIM BAY INSPECTION ATTITUDE (312,248,0)

V49 MNVR TO DOCKING ATTITUDE (180,289,0)

DOCKING ATTITUDE
VERIFY HGA

BMAG MODE (3)-ATT 1/RATE 2
CUE MSFN FOR LOGIC ARM
SECS LOGIC (BOTH)-ON(UP)
MSFN GO FOR PYRO ARM
SECS PYRO ARM (2)-ON(UP)

P47
DAC/TV-ON
LM PITCH DOWN 90°, YAW LEFT 120°
TRANSLATE TO CAPTURE LATCH
PERFORM DOCKING CHECKLIST

DOCKING

P00
V48 (61111
11111)

CMC MODE-AUTO
SC CONT-CMC
BMAG MODE (3)-RATE 2
RNDZ XPNDR-OFF
GO TO 173:35 VOL II OF FLIGHT PLAN

DOCKING CHECKLIST

AT CAPTURE

PROBE EXT/REL TB(2)-BP (IF TB NOT BP,
GO TO PG S/2-11,A)
REPORT CAPTURE TO LM
SC CONT-CMC
CMC MODE-FREE
ALLOW PROBE TO DAMP S/C MOTION(10 SEC)
WHEN WITHIN $\pm 3^\circ$ OF DOCKING ATTITUDE
PROBE RETRACT SEC-1 (PRIM-2 IF REQ)

AT DOCK LATCH

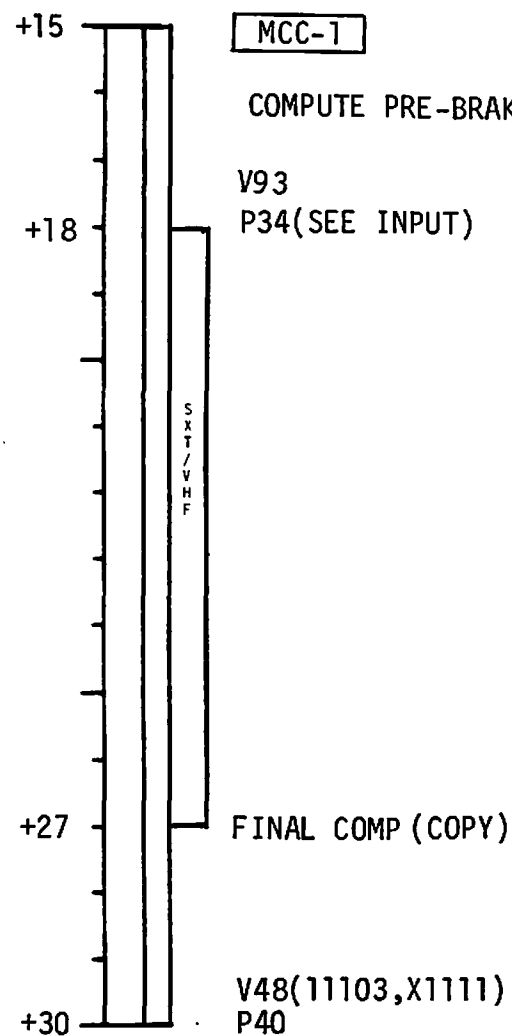
PROBE EXT/REL TB(2)-GRAY

AFTER HARD DOCK

SECS PYRO ARM (2) - SAFE
SECS LOGIC (BOTH) - OFF
CB SECS ARM (2) - OPEN
CB DOCK PROBE (2) - OPEN
THC - LOCKED
RHC - LOCKED
BMAG MODE (3) - RATE 2
PROBE EXT/REL - OFF
PROBE RETRACT (2) - OFF
EXT RUN/EVA LIGHT - OFF
EXT RNDZ LIGHT - OFF
COAS PWR - OFF
LIMIT CYCLE - ON
ATT DB - MAX
SC CONT - SCS
BMAG MODE (3) - ATT 1/RATE 2
AUTO RCS SEL B/D ROLL (4) - OFF
THC PWR - OFF
RHC PWR DIR (2) - OFF

(61111)
11111)

PRE-BRAKING SPS BURN PROCEDURES



GETI-TPI	:	:	.
ΔT: TPI-PRE-BRAKING	:	+37	00.00
GETI- PRE-BRAKING	:	:	.

TRANSFER & COMPUTATION	
OBTAIN TPF ΔV FROM N58 OF TPI SOLUTION	
TRANSFER } =	$\frac{\text{TPF } \Delta V}{2}$
	$\frac{.}{2}$
TRANSFER } =	.
LOAD INTO N55 (R3)	

PRE-BRAKING BURN COMPUTATION			
INPUT	37	COMPUTED P34 GETI	
	55	INTEG OPT +00000	ELEVATION +0000.0
FINAL COMP	37	PROCEED ON DISPLAYED GETI	
	55	INTEG OPT +00000	ELEVATION .
	58	PERILUNE ALT	P34 ΔV
	81	P34 ΔV-LV	TPF ΔV

+30

P40

SPS CHECKLIST

+37

PRE-BRAKING BURN

P79

ACQ MSFN HGA

V48(11102,X1111)

PERFORM PRE-DOCK CHECKLIST

P47 AT R=1.25 NM

SEC PRPLNT FUEL PRESS(4)-OPEN

V83E  KEY RELN83E 

UTIL PWR-ON (VERIFY)

DAC/TV-ON

LM PHOTOS WITH DAC/TV

TPF

EMS MODE-STBY

EMS FUNC-OFF

EXT RNDZ LT-OFF

GO TO 173:15 VOL II OF FLIGHT PLAN

PRE-DOCK CHECKLIST

MAN ATT(3)-RATE CMD(VERIFY)

LIMIT CYCLE-OFF(VERIFY)

ATT DB-MIN

RATE-LOW(VERIFY)

TRANS CONTR PWR-ON(UP)

ROT CONTR PWR DIRECT(BOTH)-MNA/MNB

SC CONT-CMC(VERIFY)

CMC MODE-HOLD

AUTO RCS SEL(16)-MNA/MNB(VERIFY)

CB DOCK PROBE(2)-CLOSED

PROBE RETR(2)-OFF(VERIFY)

PROBE EXTD/REL-RETR

PROBE EXTD/REL TB(2)-GRAY(VERIFY)

(IF TB NOT GRAY, GO TO PG S/2-12,E)

CB SECS LOGIC(2)-CLOSED(VERIFY)

CB SECS ARM(2)-CLOSED

EXT LIGHTS RUN/EVA-ON(UP)(VERIFY)

COAS PWR-ON(UP)(VERIFY)

BRAKING GATES

R,NM	Ṙ,FPS	RETICLE ANG,DEG	R,FT
1.50	45	.08	9000
1.00	30	.13	6000
.50	20	.26	3000
.25	10	.54	1500
.08	5	1.60	500
.05		2.70	300
.03		4.00	200
.02		8.50	100

IMU FAILURE RENDEZVOUS

WHEN IMU FAILURE KNOWN:
 PERFORM IMU FAILURE SWITCH CHECKLIST
 PERFORM GDC REFSMMAT DETERMINATION PROCEDURE(G/7-13)
 (IF REQD: PERFORM GDC REFSMMAT REALIGN(P52-OPT 4)(G/7-14)

MNVR TO TRACK/STAR FIELD ATT (180,340,0), OMNI D
 LOAD N20 AND ATT SET TW'S TO (180,340,0)
 MNVR BY NULLING ERROR NEEDLES

SYSTEMS TEST-XPNDR/C

LOAD N37

SET REFSMMAT FLAG(VERIFY)
 V25 N7E,77E,10000E,1E

IMU FAILURE SWITCH CHECKLIST

IMU POWER-OFF
 FDAI SELECT-1
 FDAI SOURCE-ATT SET
 ATT SET-GDC
 ATT DB-MIN
 RATE-LOW
 SC CONT-SCS
 BMAG MODE(3)-ATT 1/RATE 2

DIRECT ASCENT RNDZ PAD						UPDATE (IF REQ'D)					
GETI LIFT-OFF	HRS	+	0	0		+	0	0			
	MIN	+	0	0	0	+	0	0	0		
	SEC	+	0			+	0				
GETI	HRS	+	0	0		+	0	0			
TPI	MIN	+	0	0	0	+	0	0	0		
N37	SEC	+	0			+	0				

COELLIPTIC RNDZ PAD						UPDATE (IF REQ'D)					
GETI LIFT-OFF	HRS	+	0	0		+	0	0			
	MIN	+	0	0	0	+	0	0	0		
	SEC	+	0			+	0				
GETI	HRS	+	0	0		+	0	0			
CSI	MIN	+	0	0	0	+	0	0	0		
N11	SEC	+	0			+	0				
GETI	HRS	+	0	0		+	0	0			
TPI	MIN	+	0	0	0	+	0	0	0		
N37	SEC	+	0			+	0				

+55

GO/NO GO FOR LIFT-OFF
VHF VOICE CHECK
MSFN DISABLES MSFN S-BAND RELAY
GO VHF COMM

VHF ANT-RIGHT

LM LIFT-OFF

0

VHF RNG-RESET

LM INSERTION

-45

154.62
-448.0P52(OPTION 3)
RECORD STAR NO.

SHAFT

TRUNNION

P00
MSFN UPLINK:
LM S.V.

-37

P34 (ACCEPT MNKY) (BYPASS MNVR)
V57-LOAD R2=1

V
H
F

O
N
L
Y

BEFORE STEADY STATE

PRE-TPI: N49 > (+00200,+00120) REJECT/REPEAT

POST-TPI: N49 > (+00080,+00050) REJECT/REPEAT

AFTER STEADY STATE

ANYTIME: N49 > (+00030,+00020) REJECT/REPEAT

VERIFY LM TRACKER LT-ON

-31

PERFORM GDC REFSMMAT DETERMINATION PROCEDURE
V37E00E; V96E; V21 N1E,1214E,63E;
V25 N26E,13001E,3425E,30005E; V30E

SCS BURN MANEUVER

N/A SET STARS	C	S	M	B	/	O	PURPOSE
	S	P	S	G	&	N	PROP/GUID
	+			N	/	A	WT N47
R ALIGN N / A		0	0	N	/	A	P TRIM N48
P ALIGN N / A		0	0	N	/	A	Y TRIM
Y ALIGN N / A	+	0	0				HRS GETI
	+	0	0	0			MIN N33
	+	0					SEC
ULLAGE							ΔV_X N81
4 JET, 11 SEC							ΔV_Y
							ΔV_Z
	X	X	X				R
	X	X	X				P
	X	X	X				Y
ΔVC	+			N	/	A	H _A N44
				N	/	A	H _P

*IF LM BAILOUT REQ'D:
* COPY P76 DATA FROM LM

*IF CSM BAILOUT REQ'D:

*MSFN UPDATE:
* CSM BAILOUT P30 PAD
*P30
*P40; SET UP EMS
*SPS BURN CUE CARD
*CSM BAILOUT BURN
*GO TO RESCUE BOOK PG 6

P34 INPUT

37	LM GETI-TPI	:	:
55	INTEG OPT	ELEVATION	TRANSFER
	+00000	+000.00	+130.00

IMU FAILURE
RENDEZVOUS

-25

P34 (ACCEPT MNKY)(NO SXT MARKS UNTIL TPI - 17)
V25 N7E,76E,20000E,0E
V21 N1E,301E,3777E
(V86E TO REJECT MARK)

V
H
F
O
N
L
Y

-17

BEGIN SXT MARKS

V
H
F
A
N
D
S
X
T

* IF MARK REJECT BUTTON INADVERTENTLY PUSHED: *
* BEFORE PROCEEDING ON F 51 *
* V21 N1E,1336E,31264E *

VERIFY DSE TAPE MOTION(LBR/RCD/FWD/CMD RESET)

-10

P34 FINAL COMP
COMPARE SOLUTIONS

- 8

P40(35°)(180,280,0)
LOAD N18 VALUES IN N20 AND ATT SET TW'S
MNVR BY NULLING ERROR NEEDLES

- 6

SPS CHECKLIST

P00

TPI

: : .

LM +70.6,-0.5,+21.0
CSM -70.9,+0.5,-18.5
180,280,0

GROUND TPI FOR LM

						ΔV_X
						ΔV_Y
						ΔV_Z

P34 FINAL COMP

	INTEG OPT	ELEVATION §	TRANSFER §
55	+00000	.	+130.00
58	PERILUNE ALT	TPI ΔV	TPF ΔV
81	TPI ΔV -LV	.	.
84	LM TPI ΔV -LV	.	.
84	LM TPI ΔV -LV	.	.

P76

P76(P77)

P35(Accept MNKY)(5°)(180,285,0)

CALL N22; LOAD N22 VALUES IN N20 AND ATT SET TW'S
MNVR BY NULLING ERROR NEEDLES

V25 N7E,76E,20000E,0E

V21 N1E,301E,37777E

V
H
F
A
N
D
S
X
T

P35 FINAL COMP

P41(P40); LOAD N18 VALUES IN N20 AND ATT SET TW'S
MNVR BY NULLING ERROR NEEDLES

P00

MCC-1

P76(P77)

P34(Accept MNKY)(180,259,0)

V93

CALL N22; LOAD N22 VALUES IN N20 AND ATT SET TW'S
MNVR BY NULLING ERROR NEEDLES

V25 N7E,76E,20000E,0E

V21 N1E,301E, 37777E

V
H
F
A
N
D
S
X
T

P34 FINAL COMP

P41(P40); LOAD N18 VALUES IN N20 AND ATT SET TW'S
MNVR BY NULLING ERROR NEEDLES

P00

MCC-2

P35 FINAL COMP

81	MCC1 ΔV-LV	.	.
84	LM MCC1 ΔV-LV	.	.
84	LM MCC1 ΔV-LV	.	.

P76

P34 INPUT

37	GETI-MCC2	:	:
55	INTEG OPT +00000	ELEVATION \$ +000.00	TRANSFER \$ +040.00

P34 FINAL COMP

55	INTEG OPT +00000	ELEVATION \$.	TRANSFER \$ +040.00
58	PERILUNE ALT	MCC2 ΔV	TPF ΔV
81	MCC2 ΔV-LV	.	.
84	LM MCC2 ΔV-LV	.	.
84	LM MCC2 ΔV-LV	.	.

P76

+ 3

+12

+15

+18

+26

+30

P76(P77)

P79(180,289,0)

LOAD N22 VALUES IN ATT SET TW'S
MNR BY NULLING ERROR NEEDLES

ACQ MSFN HGA P -57, Y 45

PERFORM PRE-DOCK CHECKLIST

CENTER LM IN COAS
CYCLE BMAG MODE SWITCHES-RATE 2/ATT 1 RATE 2
FDAI SOURCE-GDC(VERIFY)

IF CSM ACTIVE:

AT R=1.25 NM; LOAD EMS WITH -(100+TPFΔV)
FOR DSKY DISPLAY CF VHF RANGE:
V25 N7E,75E,100E,1E; V87E; V16 N2E,3703E
SEC PRPLNT FUEL PRESS(4)-OPEN
EMS FUNC-ΔV
EMS MODE-NORMAL

TPF

LM 31.7 (TOTAL)
CSM 33.6 (TOTAL)
180,289,0

EMS MODE-STBY
EMS FUNC-OFF
EXT RNDZ LT-OFF

LM STATIONKEEP

POO

CUE MSFN FOR LOGIC ARM
SECS LOGIC(BOTH)-ON(UP)
MSFN GO FOR PYRO ARM
SECS PYRO ARM(2)-ON(UP)

LM PITCH DOWN 90°, YAW LEFT 120°

TRANSLATE TO CAPTURE LATCH
PERFORM DOCKING CHECKLIST

DOCKING

PRE-DOCK CHECKLIST

FDAI SOURCE-GDC
LIMIT CYCLE-OFF(VERIFY)
TRANS CONTR PWR-ON(UP)
ROT CONTR PWR DIRECT(BOTH)-MNA/MNB
AUTO RCS SEL(16)-MNA/MNB(VERIFY)
CB DOCK PROBE(2)-CLOSED
PROBE RETRACT(2)-OFF(VERIFY)

PROBE EXT/REL - RETR
PROBE EXT/REL TB (2) - GRAY (VERIFY)
(IF TB NOT GRAY, GO TO PG 5/2-12, E)
CB SECS LOGIC (2) - CLOSED (VERIFY)
CB SECS ARM (2) - CLOSED
EXT LIGHTS RUH/EVA - ON (UP) (VERIFY)
COAS PWR - ON (UP) (VERIFY)

BRAKING GATES

R,NM	R,FPS	RETICLE ANG,DEG	R,FT
1.50	45	.08	9000
1.00	30	.13	6000
.50	20	.26	3000
.25	10	.54	1500
.08	5	1.60	500
.05		2.70	300
.03		4.00	200
.02		8.50	100

DOCKING CHECKLIST

AT CAPTURE

PROBE EXT/REL TB (2) - BP
(IF TB NOT BP, GO TO PG 5/2-11, A)
REPORT CAPTURE TO LM
BMAG MODE (3) - RATE 2
ALLOW PROBE TO DAMP SC MOTION (10 SEC)
WHEN WITHIN ± 3° OF DOCKING ATTITUDE
PROBE RETRACT SEC - 1 (PRIM - 2 IF REQ'D)

AT DOCK LATCH

PROBE EXT/REL TB (2) - GRAY

AFTER HARD DOCK

SECS PYRO ARM (2) - SAFE
SECS LOGIC (BOTH) - OFF
CB SECS ARM (2) - OPEN
CB DOCK PROBE (2) - OPEN
THC - LOCKED
RHC - LOCKED
EXT RUH/EVA LIGHT - OFF
EXT RNDZ LIGHT - OFF
COAS PWR - OFF
LIMIT CYCLE - ON
ATT DB - MAX
BMAG MODE (3) - ATT 1/RATE 2
AUTO RCS SEL B/D ROLL (4) - OFF
THC PWR - OFF
RHC PWR DIR (2) - OFF